10. Playfair Encryption

Program

#include <stdio.h> #include <string.h> #include <ctype.h>

char matrix[5][5] = { {'M','F','H','I','K'}, {'U','N','O','P','Q'}, {'Z','V','W','X','Y'}, {'E','L','A','R','G'}, {'D','S','T','B','C'} };

void findPosition(char ch, int *row, int *col) { int i, j;

```
if (ch == 'J')
    ch = 'I';

for (i = 0; i < 5; i++)
{
    for (j = 0; j < 5; j++)
    {
        if (matrix[i][j] == ch)
        {
            *row = i;
            *col = j;
            return;
        }
    }
}


}
```

void encryptPair(char a, char b) { int r1, c1, r2, c2;

```
findPosition(a, &r1, &c1);
findPosition(b, &r2, &c2);

if (r1 == r2)
{
    printf("%c%c",
           matrix[r1][(c1 + 1) % 5],
           matrix[r2][(c2 + 1) % 5]);
```

```
}
else if (c1 == c2)
{
    printf("%c%c",
           matrix[(r1 + 1) % 5][c1],
           matrix[(r2 + 1) % 5][c2]);
}
else
{
    printf("%c%c",
           matrix[r1][c2],
           matrix[r2][c1]);
}


}
```

int main() { char text[] = "MUSTSEEYOUOVERCADOGANWESTCOMINGATONCE"; char prepared[100]; int i, j = 0;

```
/* Prepare plaintext */
for (i = 0; text[i] != '\0'; i++)
{
    if (isalpha(text[i]))
    {
        char ch = toupper(text[i]);

        if (ch == 'J')
            ch = 'I';

        prepared[j++] = ch;
    }
}

prepared[j] = '\0';

printf("Prepared Text:\n");

for (i = 0; i < j; i += 2)
{
```

```
    printf("%c%c ", prepared[i],
           (i + 1 < j) ? prepared[i + 1] : 'X');
}

printf("\n\nEncrypted Text:\n");

for (i = 0; i < j; i += 2)
{
    char a = prepared[i];
    char b;

    if (i + 1 < j)
        b = prepared[i + 1];
    else
        b = 'X';

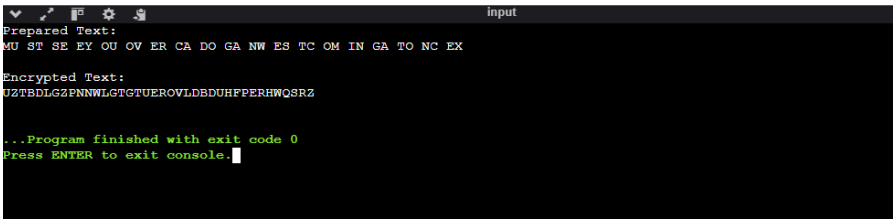
    if (a == b)
    {
        encryptPair(a, 'X');
        i--;
    }
    else
    {
        encryptPair(a, b);
    }
}

printf("\n");

return 0;


}
```

Output


input
Prepared Text:
MU ST SE EY OU OV ER CA DO GA NW ES TC OM IN GA TO NC EX
Encrypted Text:
UZTBDLGZPNNWLGTGTUEROVLDBDUHFPERHWQSRZ
...Program finished with exit code 0
Press ENTER to exit console.